

Solution to Homework 1

Guido Mazzuca

Undergraduate Exercises

1. De Moivre's Identity.

Use De Moivre's formula to prove the following identities:

$$\cos(3\theta) = \cos^3(\theta) - 3\cos(\theta)\sin^2(\theta), \quad \sin(3\theta) = 3\cos^2(\theta)\sin(\theta) - \sin^3(\theta).$$

Solution:

De Moivre's formula states that $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$. Let $n = 3$:

$$(\cos \theta + i \sin \theta)^3 = \cos(3\theta) + i \sin(3\theta)$$

Expand the left side using the binomial theorem $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$:

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3\cos^2 \theta(i \sin \theta) + 3\cos \theta(i \sin \theta)^2 + (i \sin \theta)^3$$

Substitute $i^2 = -1$ and $i^3 = -i$:

$$= \cos^3 \theta + i(3\cos^2 \theta \sin \theta) - 3\cos \theta \sin^2 \theta - i \sin^3 \theta$$

Group real and imaginary parts:

$$= (\cos^3 \theta - 3\cos \theta \sin^2 \theta) + i(3\cos^2 \theta \sin \theta - \sin^3 \theta)$$

Equating the real parts gives $\cos(3\theta) = \cos^3 \theta - 3\cos \theta \sin^2 \theta$.

Equating the imaginary parts gives $\sin(3\theta) = 3\cos^2 \theta \sin \theta - \sin^3 \theta$.

2. Sketching Sets.

Sketch the following sets in the complex plane:

$$A = \{z \in \mathbb{C} \mid |z - 1 + i| = 2\}, \quad B = \{z \in \mathbb{C} \mid \operatorname{Re}(z^2) > -2\operatorname{Im}(z) + 2\}.$$

Solution:

Set A: Rewrite the condition as $|z - (1 - i)| = 2$. This represents a circle centered at $z_0 = 1 - i$ (coordinate $(1, -1)$) with radius $R = 2$.

Set B: Let $z = x + iy$. Then $z^2 = (x^2 - y^2) + i(2xy)$. $\operatorname{Re}(z^2) = x^2 - y^2$ and $\operatorname{Im}(z) = y$. The inequality becomes:

$$x^2 - y^2 > -2y + 2$$

Rearrange terms to complete the square for y :

$$\begin{aligned}x^2 - (y^2 - 2y) &> 2 \\x^2 - (y^2 - 2y + 1 - 1) &> 2 \\x^2 - ((y - 1)^2 - 1) &> 2 \\x^2 - (y - 1)^2 &> 1\end{aligned}$$

This describes the region outside the hyperbola $x^2 - (y - 1)^2 = 1$ centered at $(0, 1)$.

3. Powers.

Compute the following powers of complex numbers:

$$(1 + i)^{13}, \quad \left(\frac{1 - i\sqrt{3}}{2} \right)^{10}.$$

Solution:

1. Let $z = 1 + i$. $|z| = \sqrt{2}$, $\text{Arg}(z) = \pi/4$.

$$(1 + i)^{13} = (\sqrt{2}e^{i\pi/4})^{13} = 2^{6.5}e^{i13\pi/4}$$

$13\pi/4 = 3\pi + \pi/4 \equiv \pi + \pi/4 = 5\pi/4$ (or simply use periodicity to get angle $-3\pi/4$).

$$= 64\sqrt{2}e^{i5\pi/4} = 64\sqrt{2}\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) = 64(-1 - i) = -64 - 64i$$

2. Let $w = \frac{1}{2} - i\frac{\sqrt{3}}{2}$. This is in Q4. $|w| = 1$, angle is $-\pi/3$.

$$w^{10} = (e^{-i\pi/3})^{10} = e^{-i10\pi/3}$$

$-10\pi/3 = -3\pi - \pi/3 = -4\pi + 2\pi/3 \equiv 2\pi/3$.

$$= \cos(2\pi/3) + i\sin(2\pi/3) = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

4. Principal Arguments.

Find the principal values of the arguments, $\text{Arg}(z)$, for the following complex numbers:

$$z_1 = -\sqrt{3} - i, \quad z_2 = (1 + i)^5, \quad z_3 = \frac{1 - i\sqrt{3}}{2}, \quad z_4 = \frac{2i}{1 + \sqrt{3}i}.$$

Solution:

(a) $z_1 = -\sqrt{3} - i$ (Q3). Ref angle $\pi/6$. $\text{Arg}(z_1) = -(\pi - \pi/6) = -5\pi/6$.

- (b) $z_2 = (1 + i)^5$. $\text{Arg}(1 + i) = \pi/4$. $\text{Arg}((1 + i)^5) = 5\pi/4$. Principal range is $(-\pi, \pi]$. $5\pi/4 - 2\pi = -3\pi/4$.
- (c) $z_3 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$ (Q4). $\text{Arg}(z_3) = -\pi/3$.
- (d) $z_4 = \frac{2i}{1+\sqrt{3}i}$. $\text{Arg}(\text{Num}) = \pi/2$. $\text{Arg}(\text{Den}) = \tan^{-1}(\sqrt{3}) = \pi/3$. $\text{Arg}(z_4) = \pi/2 - \pi/3 = \pi/6$.

5. Roots and Factorization.

Find all complex solutions to the equation:

$$z^4 + 16 = 0.$$

Use these solutions to rewrite the polynomial $P(z) = z^4 + 16$ as a product of two polynomials of degree 2 with real coefficients.

Solution:

$$z^4 = -16 = 16e^{i\pi}. \text{ Roots } z_k = 2e^{i(\pi+2\pi k)/4}.$$

$$z_0 = 2e^{i\pi/4}, z_1 = 2e^{i3\pi/4}, z_2 = 2e^{i5\pi/4}, z_3 = 2e^{i7\pi/4}$$

Grouping conjugates (z_0, z_3) and (z_1, z_2) :

$$(z - z_0)(z - \bar{z}_0) = z^2 - 2\text{Re}(z_0)z + |z_0|^2 = z^2 - 2\left(2\frac{\sqrt{2}}{2}\right)z + 4 = z^2 - 2\sqrt{2}z + 4$$

$$(z - z_1)(z - \bar{z}_1) = z^2 - 2\text{Re}(z_1)z + |z_1|^2 = z^2 - 2\left(-2\frac{\sqrt{2}}{2}\right)z + 4 = z^2 + 2\sqrt{2}z + 4$$

$$P(z) = (z^2 - 2\sqrt{2}z + 4)(z^2 + 2\sqrt{2}z + 4)$$

Graduate Exercises

1. Lagrange's Identities.

Consider the geometric sum formula for $z \neq 1$:

$$\sum_{k=0}^n z^k = \frac{1 - z^{n+1}}{1 - z}.$$

Use this result to prove Lagrange's trigonometric identities:

$$\sum_{k=0}^n \cos(k\theta) = \frac{1}{2} + \frac{\sin\left(\left(n + \frac{1}{2}\right)\theta\right)}{2\sin\left(\frac{\theta}{2}\right)}, \quad \sum_{k=0}^n \sin(k\theta) = \frac{\cos\left(\frac{\theta}{2}\right) - \cos\left(\left(n + \frac{1}{2}\right)\theta\right)}{2\sin\left(\frac{\theta}{2}\right)}.$$

Solution:

Let $S = \sum_{k=0}^n e^{ik\theta}$. Using the geometric sum formula with $z = e^{i\theta}$:

$$\begin{aligned} S &= \frac{1 - e^{i(n+1)\theta}}{1 - e^{i\theta}} = \frac{e^{i(n+1)\theta/2}(e^{-i(n+1)\theta/2} - e^{i(n+1)\theta/2})}{e^{i\theta/2}(e^{-i\theta/2} - e^{i\theta/2})} \\ &= e^{in\theta/2} \frac{-2i \sin((n+1)\theta/2)}{-2i \sin(\theta/2)} = (\cos(n\theta/2) + i \sin(n\theta/2)) \frac{\sin((n+1)\theta/2)}{\sin(\theta/2)} \end{aligned}$$

Cosine Sum (Real Part):

$$\sum \cos(k\theta) = \frac{\cos(n\theta/2) \sin((n+1)\theta/2)}{\sin(\theta/2)}$$

Using $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$:

$$Num = \frac{1}{2} [\sin((n+1)\theta) - \sin(-\theta)] = \frac{1}{2} [\sin((n+1)\theta) + \sin(\theta)]$$

$$Sum = \frac{1}{2} + \frac{\sin((n+1/2)\theta)}{2 \sin(\theta/2)}$$

Sine Sum (Imaginary Part):

$$\sum \sin(k\theta) = \frac{\sin(n\theta/2) \sin((n+1)\theta/2)}{\sin(\theta/2)}$$

Using $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$:

$$Num = \frac{1}{2} [\cos(-\theta/2) - \cos((n+1/2)\theta)]$$

$$Sum = \frac{\cos(\theta/2) - \cos((n+1/2)\theta)}{2 \sin(\theta/2)}$$

2. Taylor Series.

Using the Taylor series expansion, prove that:

$$e^{iz} = \cos(z) + i \sin(z),$$

holds for any complex number $z \in \mathbb{C}$.

Solution:

$e^{iz} = \sum_{n=0}^{\infty} \frac{(iz)^n}{n!} = 1 + iz - \frac{z^2}{2!} - i \frac{z^3}{3!} + \frac{z^4}{4!} + \dots$ Group real (even powers) and imaginary (odd powers) terms:

$$= \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \dots\right) + i \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots\right)$$

Recognizing the Maclaurin series for cosine and sine:

$$= \cos(z) + i \sin(z)$$